

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

BASIC MATH AND CALCULUS

LECTURE-6

PHYSICS

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Topics To Be Covered

- 1** Revision
- 2** Differentiation
- 3** Integration
- 4** Board and NEET PYQs
- 5** ..

Recap From 1st Lec.

Maths

- Maths Bohat Asaan hai
- number
- $+$ $-$ $\times$ $\div$
- Fraction
- Angle / Power
- Trigonometry
- Graph / Equation
- $\frac{dy}{dx}$ $\int y dx$

Fraction

- $\frac{1}{2} = 0.5$
- $\frac{1}{3} = 0.33$
- $\frac{1}{4} = 0.25$
- $\frac{1}{10} = 0.1$
- $\frac{1}{100} = 0.01$
- $5.42 \times 100 = 542$

Power

- $2^2 = 2 \times 2$
- $x^0 = 1$
- $x^a \cdot x^b = x^{a+b}$
- $(x^a)^b = x^{a \times b}$

$$\frac{1}{0} = \infty$$

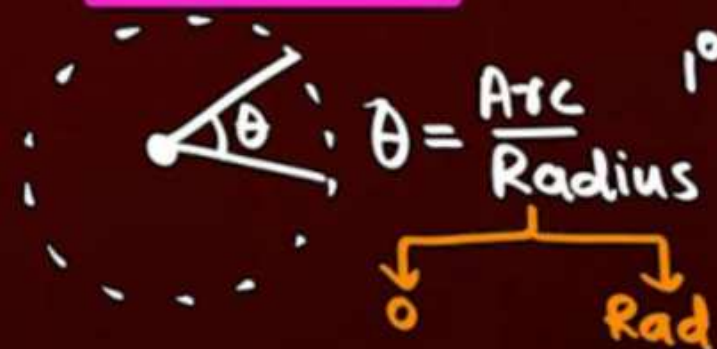
$$\frac{1}{\infty} = 0$$

$$\sqrt{2} = 2^{\frac{1}{2}}$$

$$\sqrt{2} = 1.4$$

$$\sqrt[3]{3} = 1.7$$

Angle θ



$$180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

Recap From 2nd Lec.

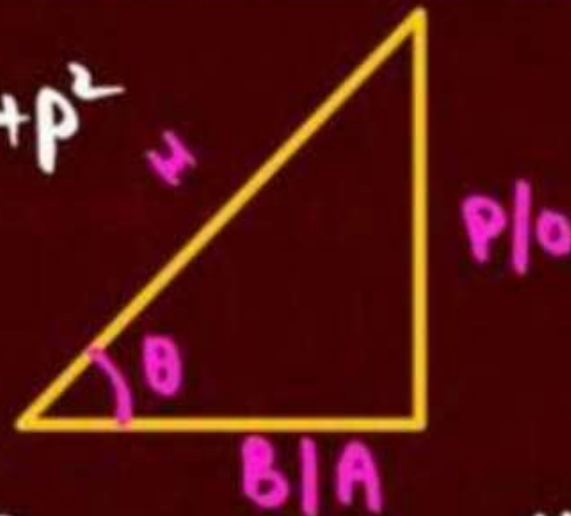
Trigonometry Table

	0°	30°	45°	60°	90°	120°	135°	150°	180°
Angle	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan \theta$	$0/1=0$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\frac{1}{0}=\infty$	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Trigonometry

Special Δ

$H^2 = B^2 + P^2$



$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$

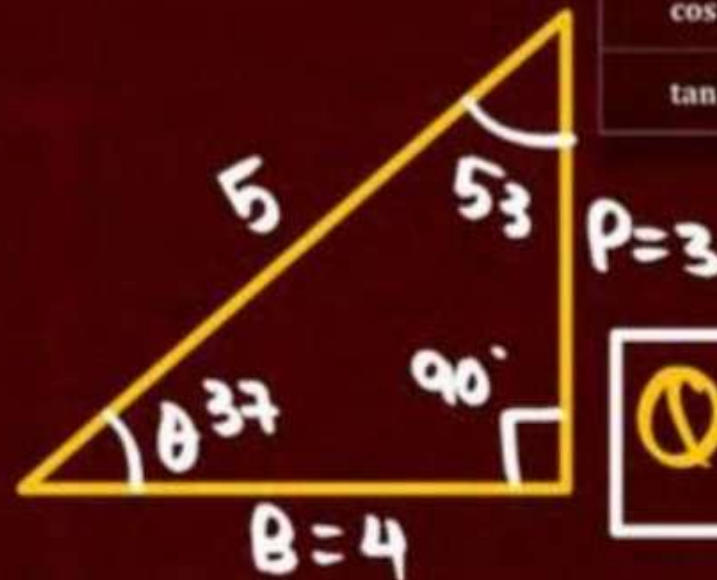
$\sec \theta = \frac{1}{\cos \theta}$

$\cot \theta = \frac{1}{\tan \theta}$

$\sin^2 \theta + \cos^2 \theta = 1$

$\sin \theta = \frac{P}{H}$
 $\cos \theta = \frac{B}{H}$
 $\tan \theta = \frac{P}{B}$

$\operatorname{cosec} \theta = \frac{H}{P}$
 $\sec \theta = \frac{H}{B}$
 $\cot \theta = \frac{B}{P}$



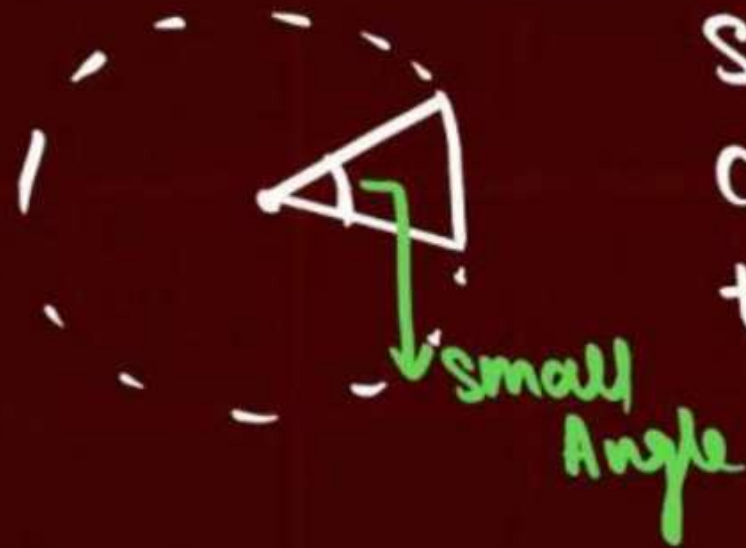
Quadrant System



Recap From 3rd Lec.

Small Angle Approx.

Equation



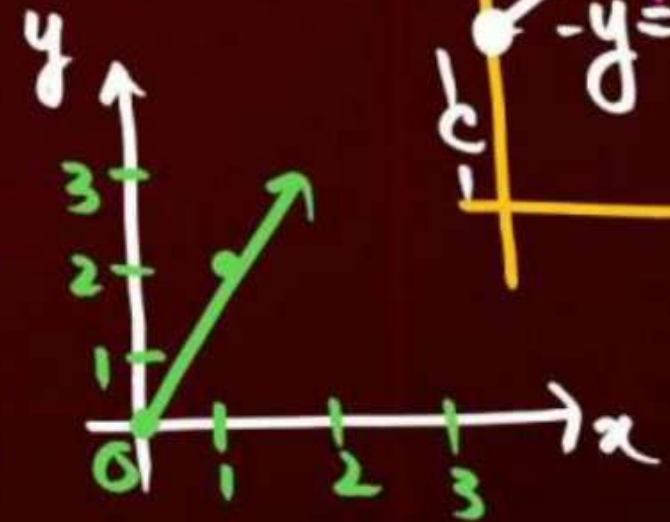
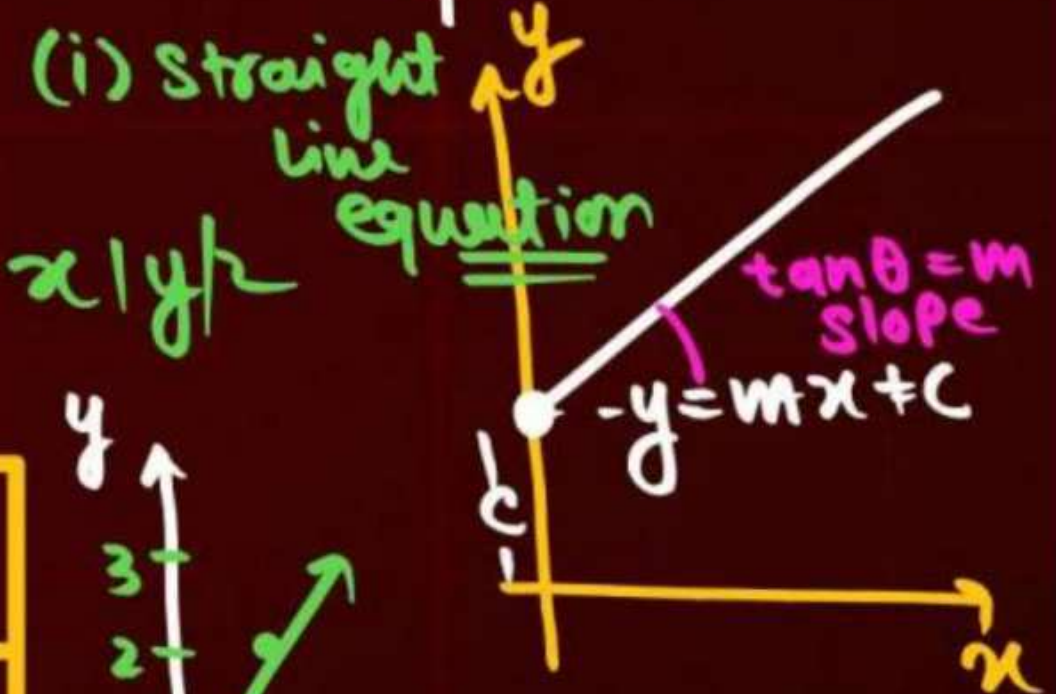
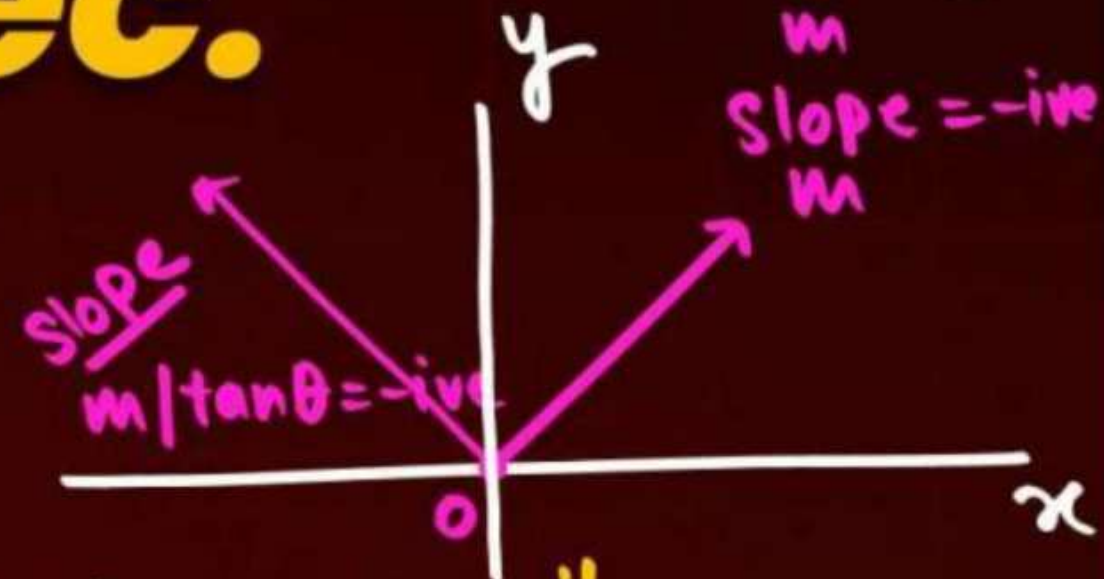
$$\begin{aligned}\sin \theta &= \theta \\ \cos \theta &= 1 \\ \tan \theta &= \theta\end{aligned}$$

English Statement

Slope $\downarrow$

$$y = \frac{dy}{dx} x = x \frac{dy}{dx}$$

x	0	1	2
y	0	2	4



Recap From 4th Lec. 3rd



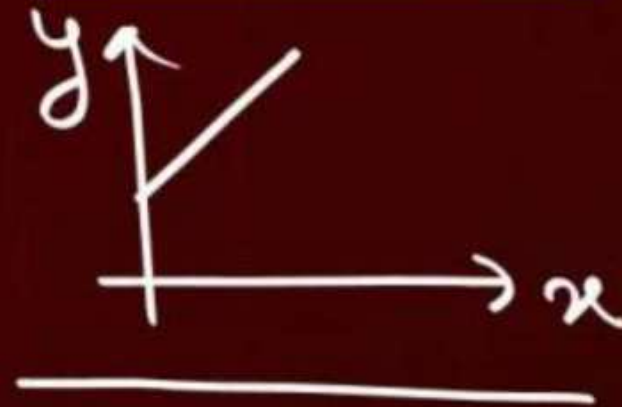
Equation

$$\underline{y = mx + c}$$

$$y = x^2$$

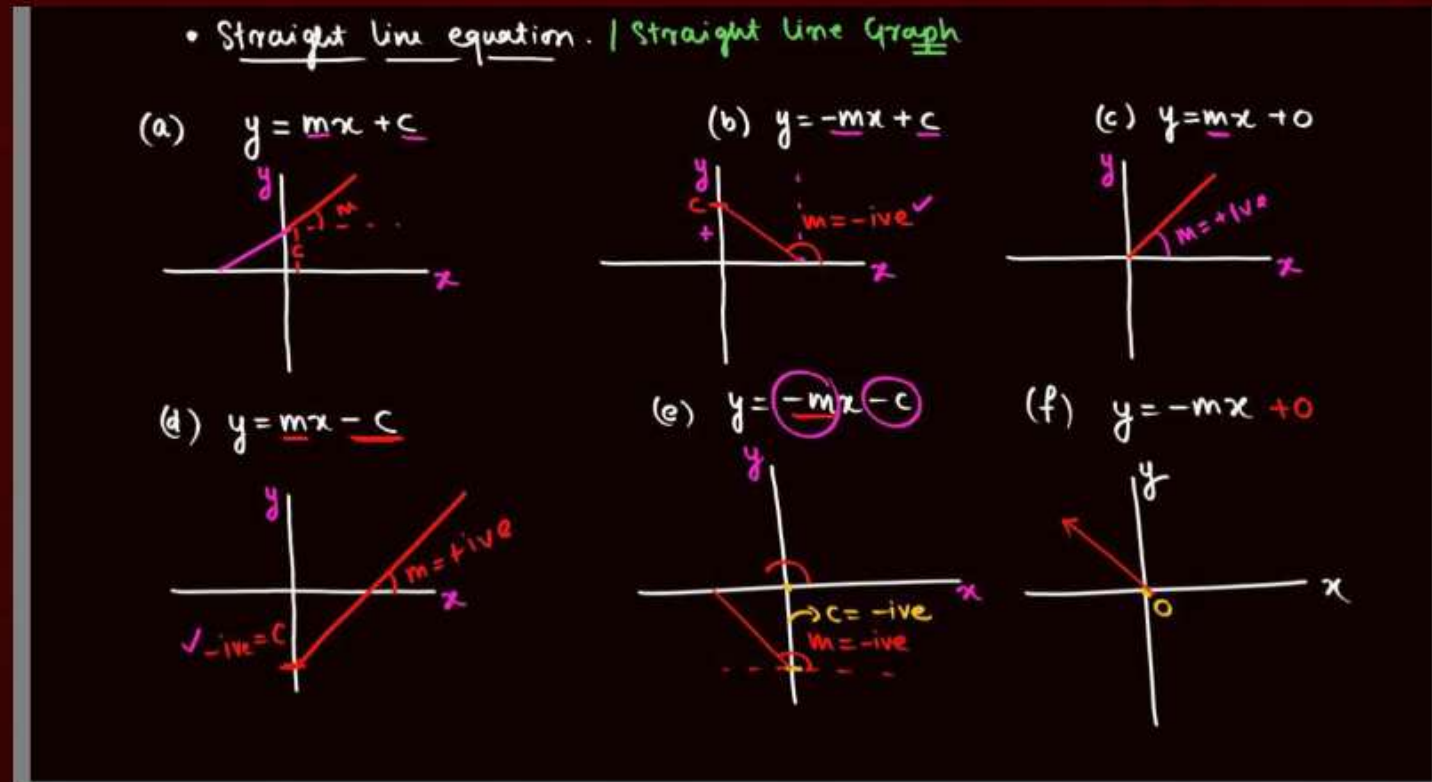
→ Table →

Graph



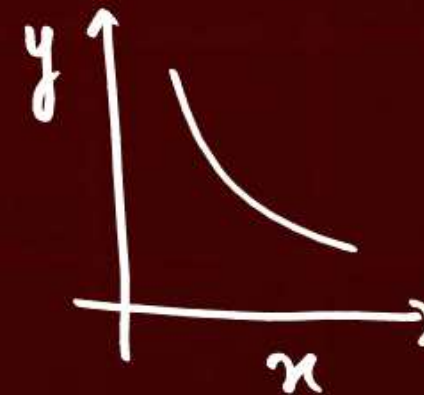
Recap From 5th Lec.

① $y = mx + c$ (Important)



② $y = x^2$

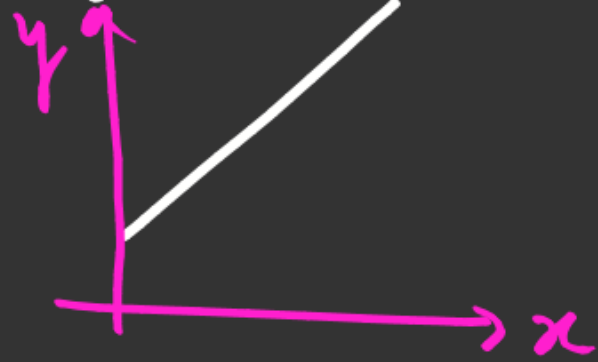
③ $y = \frac{1}{x}$



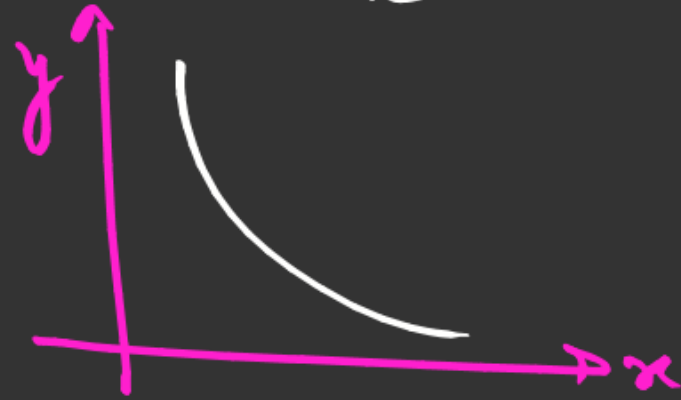
④ $x^2 + y^2 = R^2$

Lecture no. 6

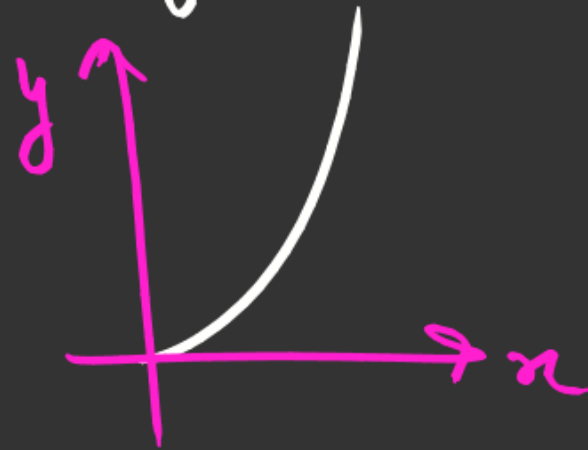
① $y = mx + c$



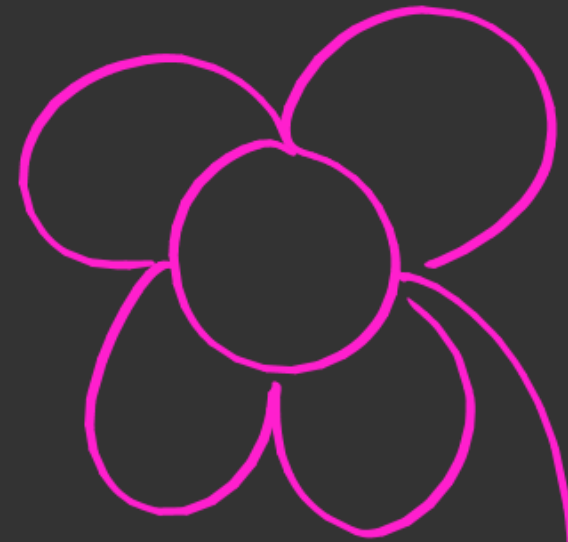
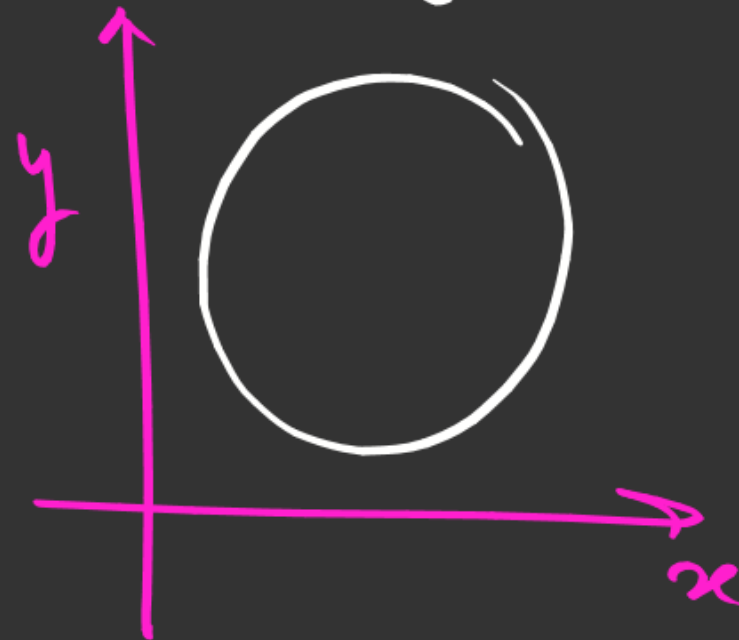
③ $y = \frac{1}{x}$



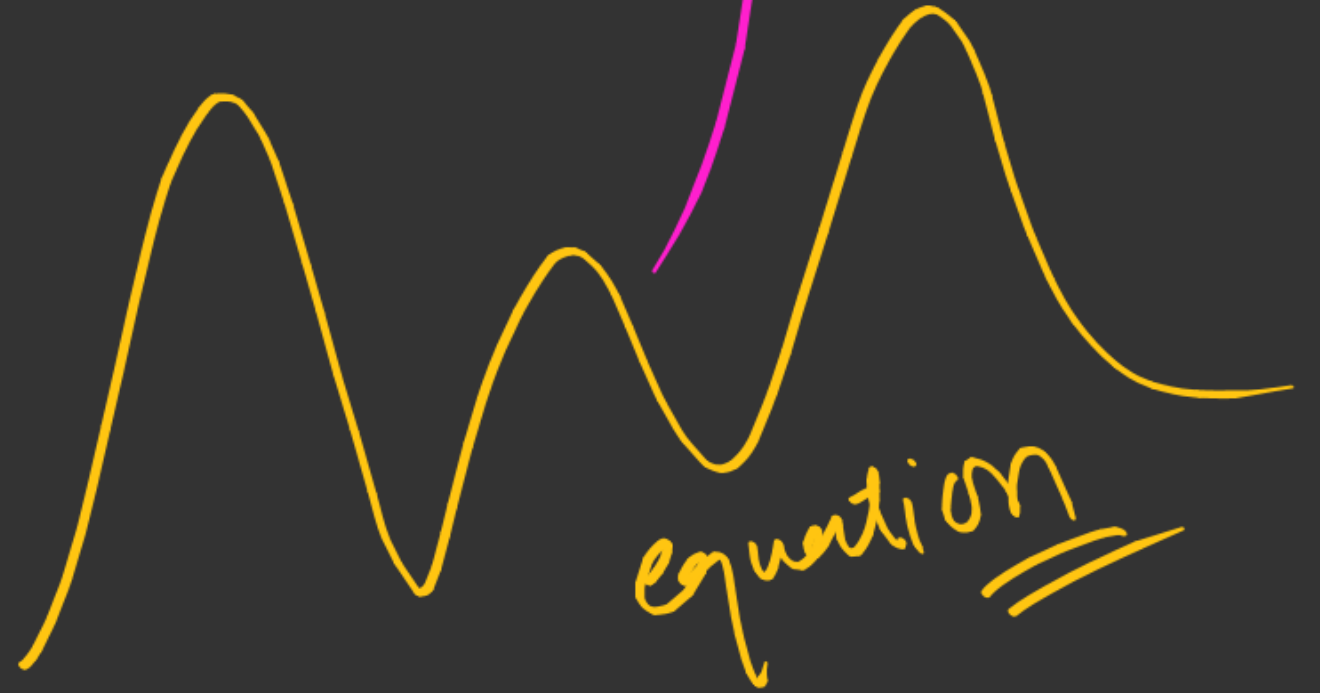
② $y = x^2$



④ $x^2 + y^2 = R^2$

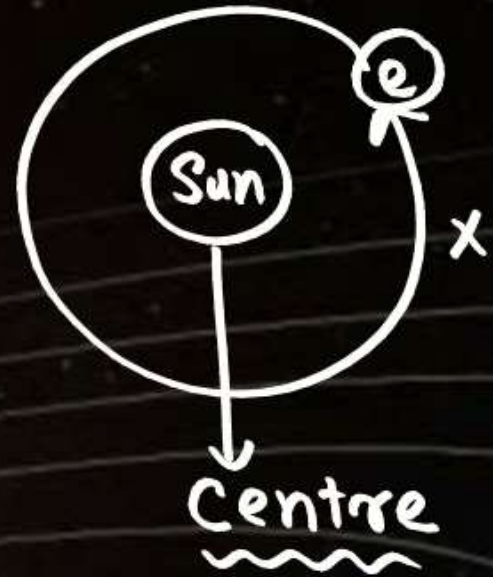


equation

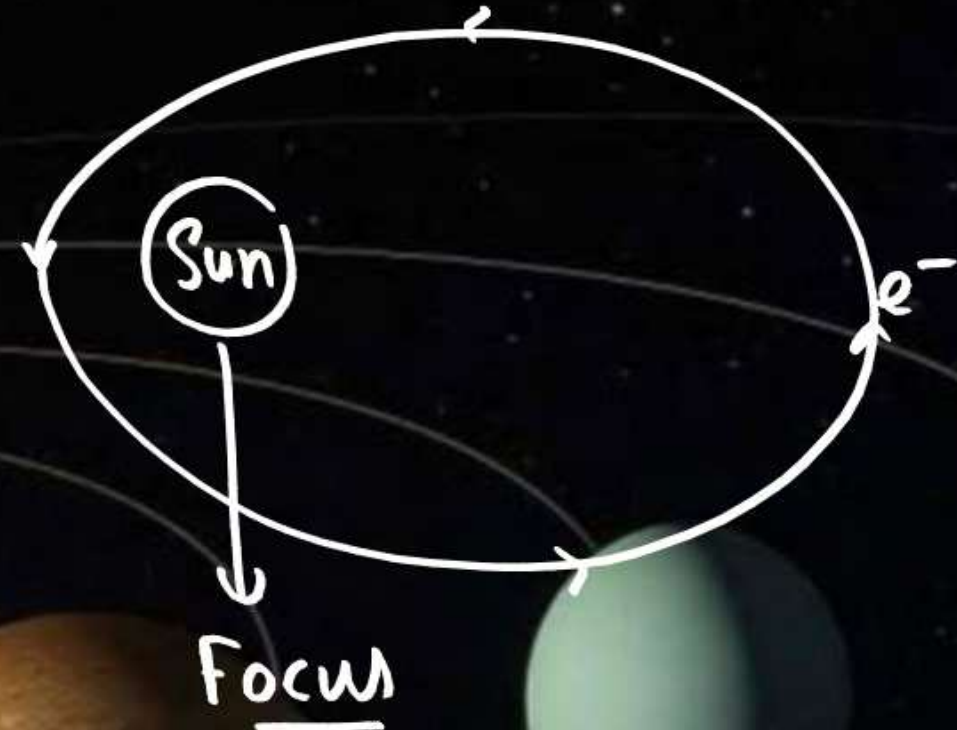


equation

Newton



Sad
chup



Calculus

Newton :- Differentiation
& integration

Differentiation

↓ equation

$$y = 2x$$

	x_1	x_2	x_3	x_4
x	<u>1</u>	<u>2</u>	3	4
y	2	4	6	8
	y_1	y_2	y_3	y_4

differentiation

$$\left(\frac{dy}{dx} \right)^x \div 0$$

↘ Bada change

$$\frac{\Delta y}{\Delta x}$$

↘ Bada change

↗ Small change in y

$$= \frac{dy}{dx}$$

↘ Small change in x

$$\frac{\text{Small change in } y}{\text{Small change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$y = 2x$

	y_1	y_2	y_3	y_4
y	2	4	6	8
x	1	2	3	4
	x_1	x_2	x_3	x_4

$y = 2x$

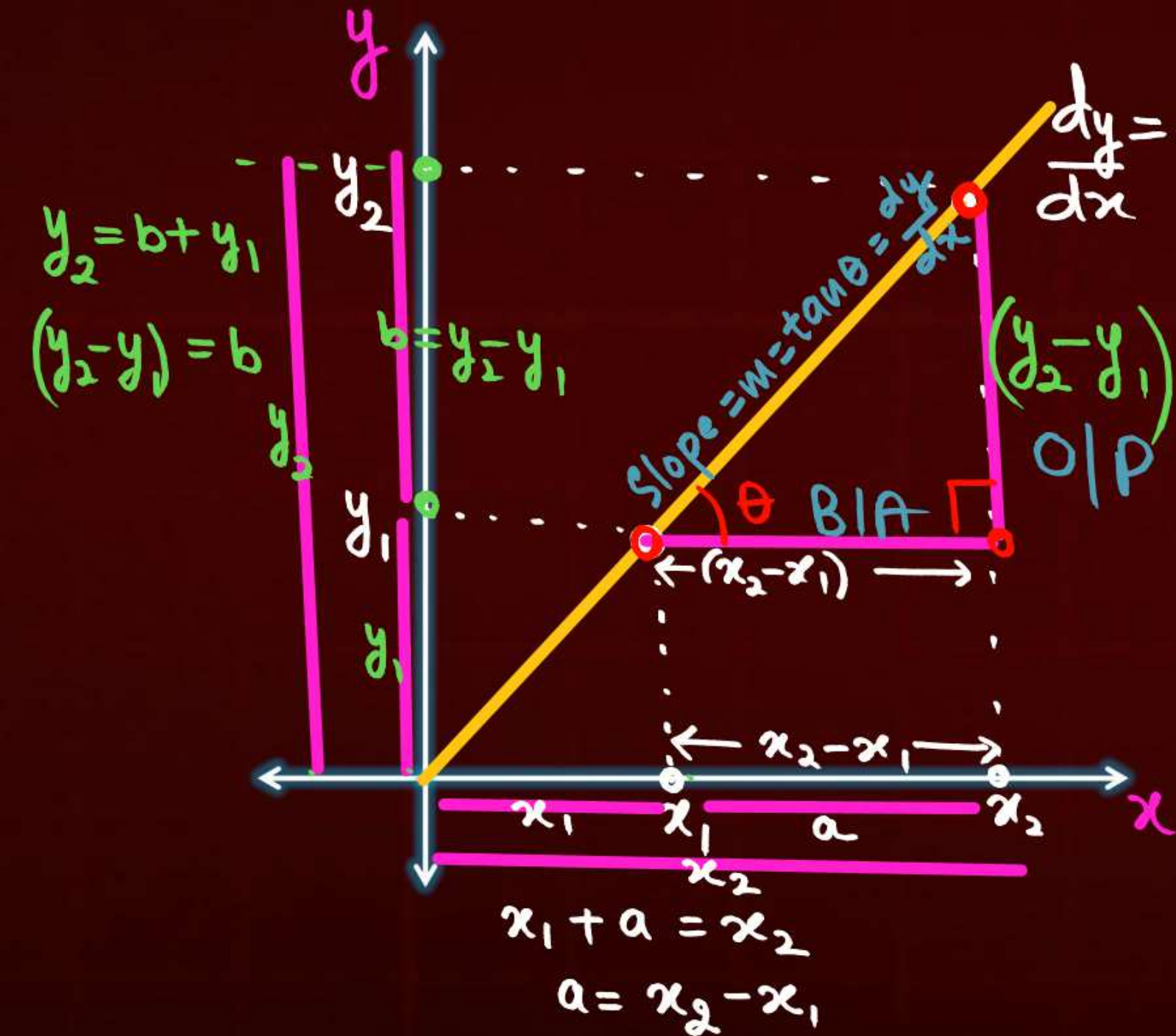
$y =$

Slope = m = $\tan \theta$ = differentiation

differentiation = $\frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 2}{2 - 1} = \frac{2}{1} = 2$

Differentiation

From Graph



$$\frac{dy}{dx} = \text{differentiation} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{P}{B} = \tan \theta \quad \checkmark$$

$$\frac{dy}{dx} = \text{differentiation} = \tan \theta = m = \text{Slope}$$

Q Board 2018 $\frac{dy}{dx} = ?$

(a) Area of graph

(b) $\frac{\text{Small change in } y}{\text{Small change in } x}$ ✓

(c) Slope of graph ✓ $m = \tan \theta = \text{Slope}$
 $= \frac{dy}{dx}$

✓ (d) b & c

Differentiation

Proof (Syllabus) 2021

Rule-1

$y = \underline{c} \rightarrow \text{constant}$

$$\frac{dy}{dx} = \frac{dc}{dx} = 0$$

Q $y = 2$ $\frac{dy}{dx} = ?$

Sol $\frac{dy}{dx} = \frac{d2}{dx} = 0$

Proof x

$$\left(\text{differentiation} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 4}{x_2 - x_1} = \frac{0}{x_2 - x_1} = 0 \right)$$

Q $\frac{d5}{dx} = ?$

(a) 1

(b) 0

(c) 5

(d) 8

Q $\frac{d6}{dx} = 0 \checkmark$

Differentiation

Rule-2

$$\underline{\underline{y = x^n}}$$

Proof $\times$

$$\underline{\underline{d x^n}} = n x^{n-1}$$

$$\frac{dy}{dx} = \frac{d x^n}{dx} = n \cdot x^{n-1}$$

Q $y = x^3$

$\frac{dy}{dx} = ?$

Sol $y = x^3$

$$\frac{dy}{dx} = \frac{dx^3}{dx} = 3x^{3-1} = \underline{\underline{3x^2}}$$

Q $y = x^4$

Sol $\frac{dy}{dx}$

$$\begin{aligned} \frac{dx^4}{dx} &= 4x^{4-1} \\ &= 4x^3 \end{aligned}$$

$$\underline{\underline{Q}} \quad y = x^4$$

$$\underline{\underline{Sol}} \quad \frac{dy}{dx} = \frac{dx^4}{dx}$$

$$= 4x^{4-1}$$

$$= \underline{4x^3}$$

$$Q \quad y = x^6$$

$$\frac{dy}{dx} = 6x^5$$

$$Q \quad y = x^7$$

$$\frac{dy}{dx} = \underline{7x^6}$$

Q $y = x^9$ (Neet-2020)

$$\frac{dy}{dx} = ?$$

Sol $\frac{dy}{dx} = \frac{d x^9}{dx} = 9x^{9-1} = \underline{\underline{9x^8}}$ ✓

Differentiation

Rule-3

$$y = e^x$$

$$\frac{dy}{dx} = \frac{de^x}{dx} = e^x$$

Q $y = e^x$

Sol $\frac{dy}{dx} = \frac{de^x}{dx} = e^x$ ✓

Differentiation

Rule-4

$$y = \ln x$$

natural log x

$$\frac{dy}{dx} = \frac{d \ln x}{dx} = \frac{1}{x}$$

$$Q_{=} y = \ln x.$$

$$Sol_{=} \frac{dy}{dx} = ?$$

$$\frac{dy}{dx} = \frac{d \ln x}{dx} = \frac{1}{x} \quad \checkmark$$

Differentiation

Rule-5 (Neet)

$$y = \underline{\underline{\sin x}}$$

$$\frac{dy}{dx} = \frac{d \sin x}{dx} = \underline{\underline{\cos x}}$$

Q $y = \sin x$

Sol $\frac{dy}{dx} = \frac{d \sin x}{dx} = \underline{\underline{\cos x}}$

Differentiation

Rule-6

$$y = \cos x$$

$$\text{Q} \quad y = \cos x$$

$$\text{Sol} \quad \frac{dy}{dx} = \frac{d(\cos x)}{dx} = -\sin x \quad \checkmark$$

$$\frac{dy}{dx} = \frac{d(\cos x)}{dx} = \underline{\underline{-\sin x}}$$

Differentiation

Rule-7

$$y = \underline{\underline{\tan x}}$$

$$\frac{dy}{dx} = \frac{d \tan x}{dx} = \underline{\underline{\sec^2 x}}$$

$$\text{Q} \quad y = \tan x$$

$$\frac{dy}{dx} = \frac{d \tan x}{dx} = \sec x$$

Differentiation

Rule-8

$$y = \operatorname{cosec} x$$

$$\frac{dy}{dx} = ?$$

Differentiation



Rule-9

$$y = \sec x.$$

$$\frac{dy}{dx} = ?$$

Differentiation

Rule-10

$$y = \cot x.$$

$$\frac{dy}{dx} = ?$$

QUESTION [JKCET 2012]

$\frac{dy}{dx}$ is?

1 Area of Graph ✗

2 $\frac{y_2 - y_1}{x_2 - x_1}$ ✓

3 Slope of the graph ✓

4 ✓ Both 2 and 3

QUESTION [NEET 2019]

$$y = \tan x$$



SUMMARY



$$\text{differentiation} = \frac{dy}{dx} = m = \text{slope} = \tan \theta$$

Rules

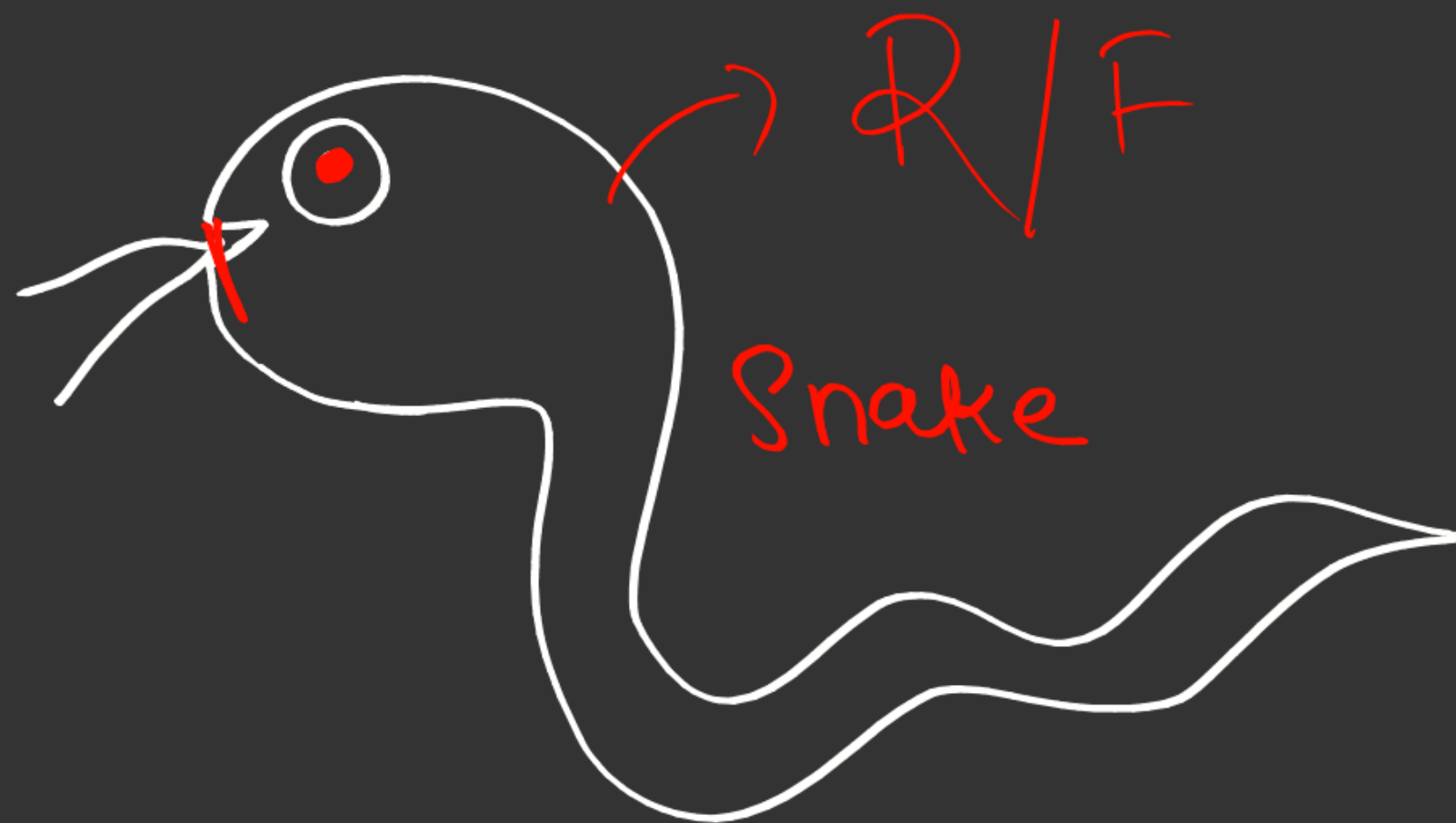
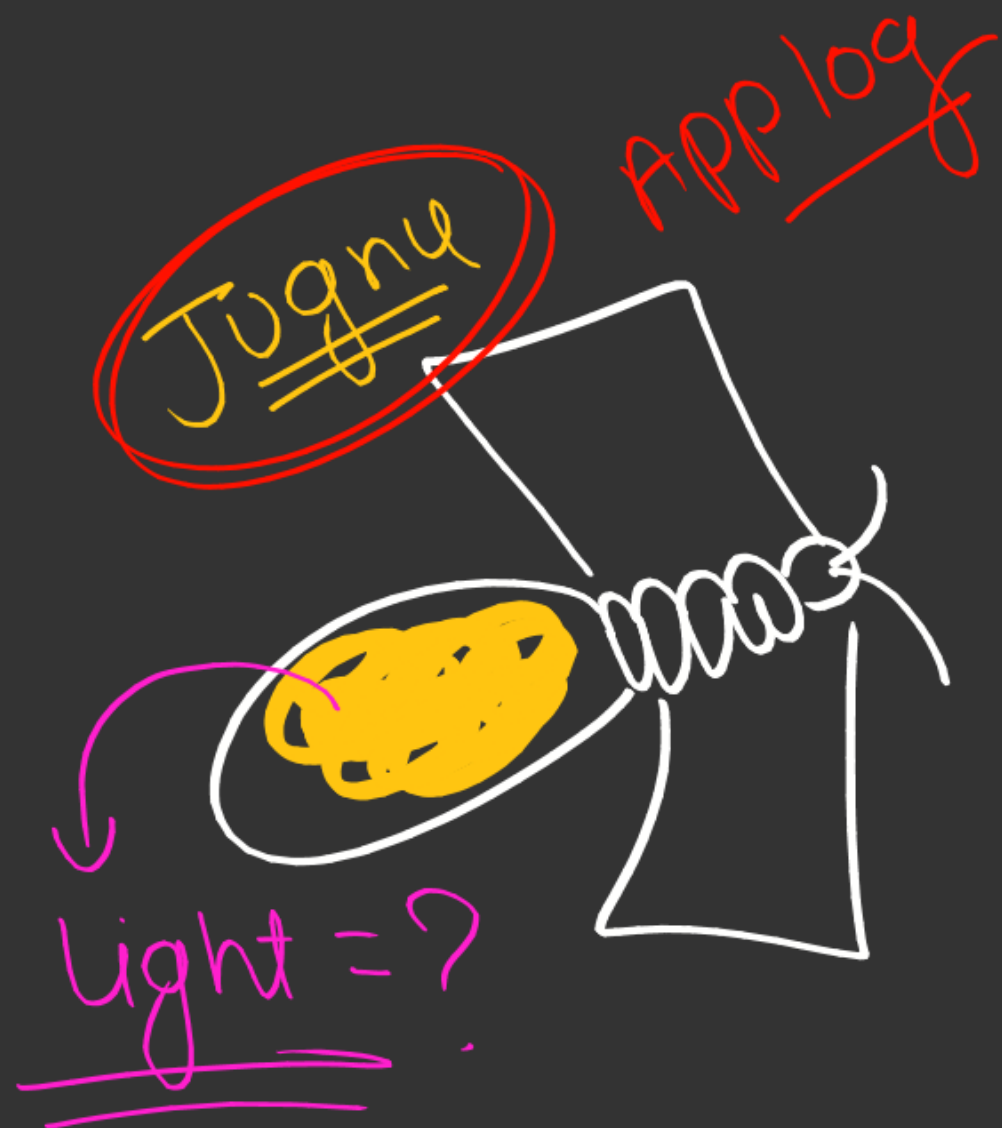
HOMEWORK



Rule 8 ✓

Rule 9 ✓

Rule 10 ✓



Thank
You